



Assessment Cover Sheet

Assessment Title	Thesis		
Due Date	19 th May 2026	Course Code	EN8914
Course Title	Final Year Project		
Internal Moderator's Name	Dr. Belal Elsayed		
External Examiner's Name			

Instructions:

1. This cover sheet must be completed (section in red below) and attached to your assessment before submission in hard copy/soft copy.
2. The time allowed for this assessment is till **19th May 2026**.
3. This assessment carries 30 % marks and assessing CILO 1, 2, 3, 4, 5 and 6.
4. The integration of AI tools and resources is encouraged to enhance learning and understanding of core theories and principles. AI-generated content, such as videos and web resources, may be utilized for brainstorming sessions and comprehensive exploration of material science concepts, provided their use adheres to ethical guidelines and academic integrity standards.
5. References consulted (if any) must be properly acknowledged and cited.

Learner ID		Date Submitted	
Learner Name			
Programme Title			
Lecturer's Name			

Learner Signature

Do not write beyond this line. For assessor use only.

Assessor's Name			
Marking Date		Maks Obtained	

Comments:

EN8914 - FINAL YEAR PROJECT: THESIS STRUCTURE AND GUIDELINES

1. Introduction and Purpose

The Final Year Project Thesis for EN8914 is the capstone academic and professional document of your engineering program at Bahrain Polytechnic. It serves as the primary assessment to demonstrate your mastery of the course and program's Learning Outcomes. This thesis is not merely a report; it is a comprehensive record that validates your ability to act as a professional engineer by designing, evaluating, and managing a complex project from conception to a validated outcome with commercial and societal relevance.

The document must systematically present the entire engineering process, from problem identification and literature review to the design, development, validation, and critical evaluation of the solution. It must justify all technical and strategic decisions, provide a robust business and sustainability analysis, and reflect on the project's societal impact, thereby demonstrating a synthesis of advanced knowledge and professional-level accountability.

2. Alignment with Learning Outcomes

Your thesis is the key piece of evidence for achieving the course's Learning Outcomes. The table below illustrates how each chapter of your thesis directly contributes to demonstrating these outcomes.

Table 1. Mapping Thesis Chapters to Learning Outcomes

Learning Outcome	Primary Chapters of Evidence	Key Evidence in Thesis
LO 1: Design an original and sustainable solution to a complex, authentic engineering problem by integrating a multidisciplinary approach that includes digital technologies and entrepreneurial insight.	Chapter 1 (Intro), Chapter 3 (Methodology), Chapter 4 (Results)	The proposed design/process/model, its integration of digital tech, and explanation of its sustainable and multidisciplinary nature.
LO 2: Critically evaluate the commercial viability, technical performance, and societal impact of a proposed solution using appropriate research methodologies, industry standards, and business metrics.	Chapter 5 (Discussion & Analysis), Chapter 6 (Conclusion)	Analysis of results against business metrics (e.g., cost-benefit), technical standards, and a dedicated section on societal impact and sustainability.
LO 3: Manage a substantial applied research project with industrial applications, synthesizing knowledge from multiple disciplines to deliver a validated outcome within constraints.	Chapter 3 (Methodology), Chapter 6 (Conclusion)	Detailed project plan within the methodology, documentation of constraints (time, budget), and the delivery of a validated outcome.

LO 4: Synthesize advanced engineering knowledge and digital tools to deliver a foundational theoretical study, coupled with at least one element of practical application, such as computer simulations or hardware development.	Chapter 2 (Literature Review), Chapter 3 (Methodology), Chapter 4 (Results)	The foundational theory in the literature review coupled with the practical application (simulation/hardware) in the methodology and results.
LO 5: Demonstrate professional accountability and ethical conduct by employing advanced communication and technical skills to synthesize and present a multidisciplinary solution to diverse stakeholders, critically evaluating its progress and impact.	Entire Thesis, Author's Declaration, Abstract, Conclusion	The clarity, structure, and professionalism of the entire document, and the critical self-evaluation of the project's impact and limitations.
LO 6: Develop a pathway to commercialize the engineering solution by developing a viable business model, evaluating market entry strategies, and demonstrating a sustainable competitive advantage.	Chapter 5 (Discussion & Analysis)	A dedicated section in the Discussion chapter on the business model, market entry strategy, and sustainable competitive advantage.

3. Thesis Structure and Formatting

The thesis must be structured into three main parts: Preliminary Pages, Main Body, and Supplementary Materials. The required order and pagination are detailed in Table 2.

Table 2. Mandatory Thesis Structure

Section	Status	Page Numbering
PRELIMINARY PAGES		
Front Cover / Title Page	Compulsory	No number
Blank Page	Compulsory	No number
Author's Declaration	Compulsory	Roman numeral (i)
Abstract	Compulsory	Roman numeral (ii)
Acknowledgements	Optional	Roman numeral (iii)
Dedication	Optional	Roman numeral (iv)
Table of Contents	Compulsory	Roman numeral (v)
List of Figures	Compulsory	Roman numeral (vi)

List of Tables	Compulsory	Roman numeral (vii)
List of Acronyms/Abbreviations	Compulsory	Roman numeral (viii)
List of Symbols	Compulsory	Roman numeral (ix)
List of Appendices	Compulsory	Roman numeral (x)
MAIN BODY		
Chapter 1: Introduction	Compulsory	Arabic numeral (1)
Chapter 2: Literature Review	Compulsory	Continues sequentially
Chapter 3: Methodology	Compulsory	Continues sequentially
Chapter 4: Results & Findings	Compulsory	Continues sequentially
Chapter 5: Discussion & Analysis	Compulsory	Continues sequentially
Chapter 6: Conclusion & Recommendations	Compulsory	Continues sequentially
References / Bibliography	Compulsory	Continues sequentially
SUPPLEMENTARY MATERIALS		
Appendices	Optional	Continues sequentially
List of Publications	Optional	Continues sequentially

Note on Chapter Structure: While Chapters 4 (Results) and 5 (Discussion) are presented as distinct sections, some projects may benefit from a combined "Results and Discussion" chapter where findings are immediately analyzed. This approach must be justified in consultation with your project supervisor.

4. Detailed Section Guidelines

5.1. Preliminary Pages

- **Front Cover / Title Page:** Must include the thesis title (concise, ≤15 words), student's name, supervisor's name, the course code (EN8914 - Final Year Project), and the Bahrain Polytechnic logo. Follow the template provided on Moodle.
- **Author's Declaration:** A signed statement confirming the work is original and your own. A template is available on Moodle.
- **Abstract:** A succinct summary (max. 250 words) of the project's purpose, methodology, key findings, and primary conclusions, including a note on commercial and societal relevance. It must stand alone as a complete overview.
- **Table of Contents, Lists of Figures/Tables/Acronyms/Symbols/Appendices:** Must accurately reflect the content, titles, and page numbers within the thesis. These are generated automatically using word processor features.

5.2. Main Body

Chapter 1: Introduction

Background and Motivation

- Start with the general context of your engineering field.
- Identify the specific gap, challenge, or real-world problem that needs solving.
- Justify it as a "complex, authentic engineering problem" that requires a sophisticated solution.

Problem Statement

- Provide a single, clear, and focused sentence or short paragraph.
- Precisely define the core problem you are addressing.

Objectives of the Project

- State the primary aim of the project.
- List 3-5 specific, measurable, and achievable (SMART) objectives that outline the key steps or deliverables.

Scope of the Project

- Define the boundaries by listing key deliverables (what you *will* do).
- State the main limitations (what you *will not* do) to keep the project focused.

Ethical Considerations (if required)

- Include only if your project involves data, people, safety, or proprietary information.
- Briefly state how you will adhere to ethical guidelines.

Significance, Sustainability, and Bahrain Perspective

- **Significance:** Explain the project's overall importance and contribution.
- **Sustainability:** Highlight its environmental, economic, and/or social benefits
- **Bahrain Perspective:** Explicitly connect the project's value to Bahrain's national goals, industry, or society

Thesis Structure

- Briefly outline the contents of the remaining chapters in the report.
- *Example:*
 - **Chapter 2: Literature Review** - Presents a critical analysis of existing research and technologies relevant to the project.
 - **Chapter 3: Methodology** - Details the materials, methods, and procedures used for the design, development, and testing phases.
 - **Chapter 4: Results and Discussion** - Presents the findings from the experiments or implementation and provides an analysis and interpretation of these results.

- **Chapter 5: Conclusion and Future Work** - Summarizes the project's achievements, reflects on the objectives, and suggests recommendations for further development.

Chapter 2: Literature Review

- Critically surveys and synthesizes existing scholarly work, technical standards, and patents relevant to the problem.
- Must include a review of relevant business models, market landscapes, and entrepreneurial approaches in the field to establish a foundation for LO6.
- Identifies gaps in current knowledge that your project aims to fill.
- Establishes the theoretical foundation for your methodology.

Chapter 3: Methodology

- Provides a detailed, replicable account of the procedures undertaken.
- Describe the research design, experimental setup, simulation parameters, fabrication processes, and/or data collection methods. Justify the selection of methods, materials, and tools.
- Details the practical application (simulation, prototype, code development, etc.) that fulfills LO4.

Chapter 4: Results & Findings

- Presents the project's outcomes clearly and objectively, without interpretation.
- Use figures, tables, and graphs to display data effectively.
- Ensure all results are directly tied to the project's objectives. This includes both technical performance results and preliminary findings related to the business case (e.g., initial cost data, market survey results).

Chapter 5: Discussion & Analysis (Enhanced)

- Interpretation of Technical Results: Relates findings back to literature and objectives, explaining their significance
- Commercial Viability and Business Model Analysis (Addresses LO6):
 - Proposed Business Model: Describe and justify a viable business model (e.g., using a Business Model Canvas framework).

- Market Entry Strategy: Analyze and evaluate potential strategies for bringing the solution to market.
- Sustainable Competitive Advantage: Articulate what makes your solution uniquely positioned to succeed in the long term.
- Financial Considerations: Discuss key business metrics, such as initial cost analysis, potential pricing, and revenue streams.
- Sustainability and Societal Impact Analysis
 - Critically evaluate the project's environmental footprint (e.g., energy use, material selection, end-of-life disposal).
 - Discuss the ethical implications and societal benefits/risks, with specific consideration to the Bahraini context.
- Critical Evaluation of Limitations and Validity: A professional assessment of the project's constraints and the reliability of the results.

Chapter 6: Conclusion & Recommendations

- Summary of Work: A concise recap of the entire project from problem definition to key findings.
- Conclusion: States the final, overarching conclusions drawn from the work, encompassing technical, commercial, and societal dimensions.
- Achievement of Objectives: Explicitly states how each initial objective (from Chapter 1) was met.
- Recommendations for Future Work: Suggests specific, actionable directions for further technical research, development, and commercialization.

References / Bibliography

- A complete list of all cited sources.
- Must consistently follow a recognized citation style (e.g., IEEE, APA) as specified by your supervisor.
- The use of reference management software (e.g., Mendeley, EndNote) is strongly encouraged.

5.3. Supplementary Materials

- **Appendices:** Include supporting information that is too detailed for the main text, such as raw data sets, extensive code listings, complex mathematical derivations, detailed circuit diagrams, survey instruments, or a completed Business Model Canvas.

5. Guidelines for Group Projects

For students undertaking a group project, a single, cohesive thesis document must be submitted for the group. The thesis must clearly demonstrate both the integrated group work and the distinct, substantial contributions of each individual member, thereby fulfilling individual assessment requirements.

5.1. Document Structure and Identification

- **Single Document:** The group will produce one thesis following all structural, formatting, and content guidelines outlined in this document.
- **Title Page:** The front cover must list the full names and student IDs of **all** group members. The primary supervisor's name shall also be included.
- **Author's Declaration:** Each group member must sign the declaration, attesting to their contribution and the originality of the collective work.

5.2. Integration of Individual Contributions

While the Abstract, Chapter 1 (Introduction), Chapter 5 (Discussion and Analysis) and Chapter 6 (Conclusion) should present a unified narrative of the group's project, the following chapters must explicitly delineate individual contributions:

Chapter 1: Introduction

The Project Objectives (Section 1.3) must include both Overall Group Objectives and Specific Individual Objectives for each member. Individual objectives should align with the broader project goals and reflect distinct, manageable components of the work.

Chapter 3: Methodology

The chapter must describe the overall project methodology. Within this description, subsections or paragraphs should be clearly attributed to specific group members, detailing the methods, tools, and processes for which they were primarily responsible (e.g., "3.2.1 Sensor Interface Design (Student A: ID 12345)", "3.3.1 Simulation Setup and Parameters (Student B: ID 67890)").

Chapter 4: Results & Findings

Presentation of results should be organized to reflect the individual contributions outlined in the Methodology. Figures, tables, and data sets should be attributed to the responsible member where appropriate (e.g., "Table 4.1: Performance metrics for the control algorithm developed by Student A").

5.3. General Principle

The thesis must read as a coherent, integrated report while providing a transparent audit trail that allows assessors to identify, evaluate, and grade the unique work of each student. The seamless integration of individual efforts into a unified solution is a key demonstration of professional engineering practice.

EN8914-Final Year Project Thesis - Marking Rubric (100 Marks)

Student Name and ID: _____

Supervisor/Examiner Name: _____

Total Mark: -----/100

CHAPTER & CRITERIA	POOR (0-59%)	GOOD (60-69%)	VERY GOOD (70-84%)	EXCELLENT (85-100%)	MARKS
1. INTRODUCTION & PROJECT DEFINITION					/10
BACKGROUND AND MOTIVATION	Context is missing or irrelevant. Problem is not justified as complex or authentic.	Context is vague. Problem is simple or poorly justified.	Context is clear. Problem is relevant and justified as a complex engineering problem.	Context is expertly established. Problem is a compelling, well-justified "complex, authentic engineering problem."	/2.5
PROBLEM STATEMENT	Problem statement is missing or irrelevant.	Statement is unclear, too broad, or not focused.	Statement is clear, specific, and addresses a core issue.	Statement is exceptionally clear, focused, and precisely defines the core engineering problem.	/2.5
PROJECT OBJECTIVES	Objectives are missing or do not address the problem.	Objectives are vague, not measurable, or not achievable.	Objectives are mostly SMART (Specific, Measurable, Achievable, Relevant, Time-bound).	All objectives are perfectly SMART and directly target the problem statement.	/2.5
PROJECT SCOPE AND LIMITATIONS	Scope is not defined.	Scope is poorly defined with unrealistic or missing in/out-of-scope items.	Scope is defined with clear in-scope and out-of-scope items.	Scope is precisely defined with realistic and insightful boundaries that perfectly manage project focus.	/1.5
SIGNIFICANCE, SUSTAINABILITY & BAHRAIN PERSPECTIVE	Significance, sustainability, and Bahrain perspective are missing or irrelevant.	Significance is weak. Sustainability and Bahrain perspective are generic or tenuous.	Significance is clear. Sustainability and a relevant Bahrain perspective are present.	Significance is powerfully argued. Sustainability is critically analyzed. Bahrain perspective is specific and linked to national goals.	/1.0
2. LITERATURE REVIEW & THEORETICAL FOUNDATION					/15
BREADTH, DEPTH & CRITICAL ANALYSIS	Sources are insufficient or irrelevant. Purely descriptive. No research gap identified.	Limited range of sources. Predominantly descriptive with little critical analysis. Gap is vague.	Good range of sources. Some critical analysis is present. A research gap is clearly identified.	Comprehensive review. Sources are highly relevant and critically evaluated. A clear research gap is compellingly identified.	/7.5
SYNTHESIS & BUSINESS CONTEXT	Fails to synthesize information. No business or market context is provided.	Lists technical information with minimal synthesis. Lacks a business or market perspective.	Synthesizes technical information and includes some review of business models or market context.	Expertly synthesizes technical and business/entrepreneurial literature to build a strong, multi-faceted foundation.	/5.0

REFERENCING	Pervasive errors in citation and referencing.	Inconsistent or incorrect citation style.	Consistent citation and referencing with minor errors.	Flawless citation and referencing in the required style.	/2.5
3. METHODOLOGY & PRACTICAL APPLICATION					/20
METHODOLOGY DESCRIPTION & JUSTIFICATION	Methods are missing, irrelevant, or cannot be replicated.	Methods are unclear or lack detail. Justification for their selection is weak.	Methods are described clearly and are mostly replicable. Selection is justified.	Methods are described in replicable detail. Selection is expertly justified against literature. Integrates digital tech/multidisciplinary approach clearly.	/10
PRACTICAL APPLICATION (SIMULATION/HARDWARE)	No meaningful practical application is evident.	The practical application is simplistic, poorly executed, or misaligned with the objectives.	A clear and appropriate practical application is described and implemented.	The practical application is complex, well-designed, and perfectly suited to address the project objectives.	/10
4. RESULTS & FINDINGS					/10
PRESENTATION & CLARITY	Data is poorly presented, inaccessible, or missing.	Data presentation is confusing, cluttered, or lacks appropriate visuals.	Data is presented clearly and objectively using appropriate figures and tables.	Data is presented objectively with exceptional clarity using professional-quality figures, tables, and graphs.	/5.0
LINK TO OBJECTIVES	No link is made to the project's objectives.	The link between results and objectives is weak or unclear.	Most results are clearly linked to the project's objectives.	All results are explicitly and directly tied to the project's initial SMART objectives.	/5.0
5. DISCUSSION, ANALYSIS & COMMERCIALIZATION					/30
TECHNICAL DISCUSSION & CRITICAL EVALUATION	No meaningful discussion or critical evaluation.	Discussion is superficial or merely restates results. Limited critical evaluation of limitations.	Results are discussed and compared to expectations/literature. Limitations are acknowledged.	Results are expertly interpreted and critically evaluated against literature and standards. Limitations are insightful and honest.	/10
SUSTAINABILITY & SOCIETAL IMPACT	Little to no discussion of sustainability or societal impact.	Sustainability and impact are mentioned but not critically analyzed. Bahrain perspective is weak.	Covers key sustainability aspects and societal impact, with a clear Bahrain perspective.	Comprehensive and critical evaluation of environmental, economic, and social impact, with a sophisticated Bahrain-specific analysis.	/10
BUSINESS MODEL & COMMERCIAL VIABILITY	Commercialization aspects are missing or entirely implausible.	Business model is vague. Commercial discussion is superficial or unrealistic.	A clear business model is proposed. Market entry strategy and competitive advantage are discussed.	A viable, well-justified business model is presented. Market entry strategy and sustainable competitive advantage are compelling and realistic.	/10
6. CONCLUSION & SYNTHESIS					/10
SUMMARY & ACHIEVEMENT OF OBJECTIVES	No meaningful summary. Fails to address initial objectives.	Summary is vague or incomplete. Link to objective achievement is unclear.	Clearly summarizes the project. Explicitly states the achievement of most objectives.	Concisely and powerfully summarizes the work. Explicitly and accurately states how each objective was met (or not).	/5.0

CONCLUSIONS & FUTURE WORK	Conclusions are missing or incorrect. Future work is not provided.	Conclusions are weak or unsupported. Future work is generic.	Conclusions are supported by the work. Future work is relevant.	Conclusions are insightful and well-supported. Recommendations for future work are specific, actionable, and cover technical & commercial aspects.	/5.0
7. PROFESSIONALISM, STRUCTURE & COMMUNICATION					/5
THESIS STRUCTURE, LANGUAGE & FORMATTING	Poorly structured and written. Difficult to follow. Does not adhere to guidelines.	Deviates from required structure. Writing has significant errors affecting readability.	Follows required structure. Writing is clear with minor grammatical or formatting errors.	Flawless adherence to structure and formatting guidelines. Professional, clear, and concise writing with no errors.	/5.0
FINAL TOTAL					/100